

Thm. Let  $\{a_n\}$  be a series, and take  $R = (\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}})^{-1}$ .

Then, the series  $\sum_{n=0}^{\infty} a_n x^n$  converges absolutely if  $|x| < R$ ,  
diverges if  $|x| > R$ .

Therefore, we can define  $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$ .

For any  $0 < \epsilon < R$ ,  $f$  converges uniformly on  $[-R+\epsilon, R-\epsilon]$  and  
 $f$  is differentiable on  $(-R, R)$ , with

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

proof. The first assertion is given by the root test.

And, the second assertion is given by the Weierstrass-M test, specifically,  
for all  $n \in \mathbb{N}$  and  $x \in [-R+\epsilon, R-\epsilon]$ ,  $|a_n x^n| = |a_n| |x|^n \leq |a_n| \cdot (R-\epsilon)^n$ , and

$$\sum_{n=0}^{\infty} |a_n| \cdot (R-\epsilon)^n$$

Revised in the next page.

converges, by the first. To show the third assertion, observe that

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}$$

Because: since  $x_n = n^{\frac{1}{n}} - 1$  satisfies  $(1+x_n)^n = n \geq \frac{n(n-1)}{2} \cdot x_n^2$ , thus

$$x_n^2 \leq \frac{2}{n-1} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus  $n^{\frac{1}{n}} \rightarrow 1$  as  $n \rightarrow \infty$ . Given  $\epsilon > 0$ , there is  $N_1 \in \mathbb{N}$  s.t

$$n \geq N_1 \Rightarrow |a_n|^{\frac{1}{n}} < \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \frac{\epsilon}{2}$$

Meanwhile, since  $n^{\frac{1}{n}} \rightarrow 1$  as  $n \rightarrow \infty$ , take  $N_2 \in \mathbb{N}$  s.t

$$n \geq N_2 \Rightarrow n^{\frac{1}{n}} < 1 + r, \text{ where } r = \frac{A + \epsilon}{A + \frac{\epsilon}{2}} - 1, \quad A = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

Then,  
 $n \geq \max\{N_1, N_2\} \Rightarrow n^{\frac{1}{n}} \cdot |a_n|^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \epsilon.$

Moreover, there are infinitely many  $n \in \mathbb{N}$  s.t  $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} - \frac{\epsilon}{2} < |a_n|^{\frac{1}{n}}$ ,

we can choose similarly, 'infinitely many  $n \in \mathbb{N}$  s.t  $1 - t < n^{\frac{1}{n}}$ ,  $t = 1 - \frac{A - \epsilon}{A - \frac{\epsilon}{2}}$

Now proved. Using this,  $\sum n a_n x^{n-1}$  converges uniformly in  $(-R, R)$ ,

we obtain the result.

Thm. Let  $\{a_n\}$  be a sequence in  $\mathbb{R}$  with  $R = \left(\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}\right)^{-1}$ .

Then, the power series

$$\sum_{n=0}^{\infty} a_n x^n \quad \begin{array}{l} \text{Converges for all } |x| < R, \\ \text{diverges for all } |x| > R. \end{array} \quad \text{By root test.}$$

Therefore, we can define

$$f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n.$$

The function  $f$  satisfies:

1) For any  $0 < \epsilon < R$ ,  $f$  is uniformly continuous on  $[-R+\epsilon, R-\epsilon]$

2)  $f \in C^\infty(-R, R)$  with  $f'(x) = \sum_{n=1}^{\infty} n \cdot a_n x^{n-1}$ .

Proof. Let  $x \in (-R, R)$  be given. Then, for each  $n \in \mathbb{N}$ ,

$$\sqrt[n]{|a_n \cdot x^n|} = \sqrt[n]{|a_n| \cdot |x|^n} = |a_n|^{\frac{1}{n}} \cdot |x|$$

Therefore, by the root test,  $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \cdot |x| = |x| \cdot \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} < R \cdot R^{-1} = 1$ ,  
the given series converges. ( $\sqrt[n]{|b_n|} < 1 \Rightarrow a_n < \beta^n$  for some  $\beta < 1$ .)

To show 1), let  $0 < \epsilon < R$  be given. For each  $n \in \mathbb{N}$ , and  $x \in [-R+\epsilon, R-\epsilon]$ ,

$$|a_n x^n| = |a_n| \cdot |x|^n \leq |a_n| \cdot |R-\epsilon|^n \quad \text{and} \quad \sum_{n=0}^{\infty} |a_n| \cdot |R-\epsilon|^n < \infty,$$

therefore by the Weierstrass M-test,  $\sum a_n x^n$  converges uniformly to  $f$ ,  
so the  $f$  is continuous on the compact set  $[-R+\epsilon, R-\epsilon]$ , so uniformly conti.

To show 2), put  $f_N(x) = \sum_{n=0}^N a_n x^n$ . Then,  $f'_N(x) = \sum_{n=1}^N n a_n x^{n-1}$ .

Since  $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}$ ,  $\lim_{N \rightarrow \infty} f'_N(x)$  converges in  $(-R, R)$ .

Moreover,  $f_N$  converges to  $f$  in  $(-R, R)$  and

for any  $0 < \epsilon < R$ ,  $f'_N$  converges uniformly, because given  $x \in [-R+\epsilon, R-\epsilon]$ ,

$$|n \cdot a_n \cdot x^{n-1}| \leq n \cdot a_n \cdot |R-\epsilon|^{n-1} \quad \text{and} \quad \sum_{n=1}^{\infty} n a_n |R-\epsilon|^{n-1} \text{ converges by the root test,}$$

therefore the Weierstrass M-test gives  $f'_N$  converges uniformly.

Now, given  $x \in (-R, R)$ , we can take  $\epsilon > 0$  such that  $x < R-\epsilon < R$ , thus

$$\lim_{N \rightarrow \infty} f'_N(x) = \sum_{n=1}^{\infty} n \cdot a_n \cdot x^{n-1} = f'(x). \quad \text{by the theorem for differentiation}$$

Thm. Suppose that  $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$  is well-defined.

Given  $c \in (-R, R)$ , the representation

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \quad (|x-c| < R-|c|)$$

is valid.

Proof. Given  $c \in (-R, R)$

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot x^n = \sum_{n=0}^{\infty} a_n \cdot [x-c+c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^n \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$$

Meanwhile, since  $|x-c| + |c| < R$ ,

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=0}^n |a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m| &= \sum_{n=0}^{\infty} \sum_{m=0}^n |a_n| \cdot \binom{n}{m} \cdot |c|^{n-m} \cdot |x-c|^m \\ &= \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty. \quad \dots (*) \end{aligned}$$

Put 
$$A_{nm} = \begin{cases} a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m & \text{if } n \geq m \\ 0 & \text{if } n < m. \end{cases}$$
 Then,

$\sum_{m=0}^{\infty} |A_{nm}|$  converges for all  $n \in \mathbb{N}$ , (because for fixed  $n$ ,  $\sum_{m=0}^{\infty} |A_{nm}|$  is just finite sum,

actually,  $\sum_{m=0}^{\infty} |A_{nm}| = \sum_{m=0}^n |A_{nm}|$ .) and moreover,  $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} |A_{nm}|$  converges by (\*). Now

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \sum_{n=m}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \left[ \sum_{n=m}^{\infty} \binom{n}{m} \cdot a_n \cdot c^{n-m} \right] \cdot (x-c)^m$$

Finally,  $\frac{1}{m!} f^{(m)}(x) = \sum_{n=m}^{\infty} \binom{n}{m} \cdot a_n \cdot x^{n-m}$  gives the result, just substituting  $x=c$ .  $\square$

★ This Proof is revised at 2026.8.2.

Thm. Suppose that  $f: (-R, R) \rightarrow \mathbb{R} : x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$  is well-defined.

Given  $c \in (-R, R)$ , the equality holds:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \quad (x \in \mathbb{B}_{R-|c|}(c)).$$

Proof. Observe that

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot [(x-c) + c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^n \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m.$$

And, since  $x \in \mathbb{B}_{R-|c|}(c) \Leftrightarrow |x-c| + |c| < R$ ,

$$\sum_{n=0}^{\infty} \sum_{m=0}^n |a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m| = \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty.$$

So,  $\sum_{m=0}^{\infty} |A_{nm}|$  converges for each  $n$ , where  $A_{nm} = a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$ .

We can change the limits

$$f(x) = \sum_{m=0}^{\infty} \left( \sum_{n=m}^{\infty} \binom{n}{m} a_n \cdot c^{n-m} \right) (x-c)^m = \sum_{m=0}^{\infty} \frac{f^{(m)}(c)}{m!} \cdot (x-c)^m.$$

Thm. Given series  $\sum a_n$  and  $\sum b_n$ , define  $C_n = \sum_{k=0}^n a_k \cdot b_{n-k}$ .

If  $\sum_{n=0}^{\infty} a_n$  Converges absolutely and  $\sum_{n=0}^{\infty} b_n$  Converges, then

$$\sum_{n=0}^{\infty} C_n = \left( \sum_{n=0}^{\infty} a_n \right) \cdot \left( \sum_{n=0}^{\infty} b_n \right).$$

Proof. Denote

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad B = \sum_{n=0}^{\infty} b_n \quad \text{and} \quad \beta_n = B_n - B$$

Then,

$$\begin{aligned} \sum_{k=0}^n a_k &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \dots + (a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \dots + a_n B_0 \\ &= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \dots + a_n (B + \beta_0) \\ &= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0. \end{aligned}$$

Now, enough to show that

$$\gamma_n = a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0$$

Converges to zero.

Let  $\epsilon > 0$  be given. Since  $\beta_n \rightarrow 0$  as  $n \rightarrow \infty$ , there exists  $N \in \mathbb{N}$  s.t.  $n \geq N \Rightarrow |\beta_n| \leq \epsilon$ .

Now, for  $n \geq N$ ,

$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N+1} + \dots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot (|a_{N+1}| + \dots + |a_0|) \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot \sum_{k=0}^{\infty} |a_k| \quad \leftarrow \end{aligned}$$

Using this, Converges absolutely.

Take  $\limsup$ , then

$$\begin{aligned} \limsup_{n \rightarrow \infty} |\gamma_n| &\leq \limsup_{n \rightarrow \infty} |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \sum_{k=0}^{\infty} |a_k| \\ &= \epsilon \cdot \sum_{k=0}^{\infty} |a_k|. \end{aligned}$$

Consequently,  $\gamma_n \rightarrow 0$  as  $n \rightarrow \infty$ , so that we obtain the result.

Abel.

Thm. Let  $f: [-1, 1) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$  be defined.

If  $\sum_{n=0}^{\infty} a_n$  converges, then  $\lim_{x \rightarrow 1^-} f(x) = \sum_{n=0}^{\infty} a_n$ .

Proof. Put  $A_n = \sum_{k=0}^n a_k$ , and  $A = \lim_{n \rightarrow \infty} A_n = \sum_{n=0}^{\infty} a_n$ . As a Cauchy product,

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} A_n \cdot x^n, \text{ so } f(x) = (1-x) \cdot \sum_{n=0}^{\infty} A_n x^n \quad (-1 < x < 1)$$

and since  $(1-x) \cdot \sum_{n=0}^{\infty} x^n = 1$ , so

$$f(x) - A = (1-x) \sum_{n=0}^{\infty} A_n x^n - A \cdot (1-x) \sum_{n=0}^{\infty} x^n$$

$$= (1-x) \left( \sum_{n=0}^{\infty} A_n x^n - A \cdot \sum_{n=0}^{\infty} x^n \right)$$

$$= (1-x) \cdot \left( \sum_{n=0}^{\infty} (A_n - A) \cdot x^n \right), \quad (-1 < x < 1).$$

Given  $\varepsilon > 0$ , take  $N \in \mathbb{N}$  s.t.  $n \geq N \Rightarrow |A_n - A| < \varepsilon$ , and put  $M = \max \{|A_n - A| \mid n = 0, \dots, N\}$ .

$$\text{Now, } |f(x) - A| \leq \left| (1-x) \cdot \sum_{n=0}^N (A_n - A) x^n \right| + \left| (1-x) \cdot \sum_{n=N}^{\infty} (A_n - A) x^n \right|$$

$$\leq (1-x) \cdot MN + \varepsilon$$

Therefore,  $\lim_{x \rightarrow 1^-} f(x) = A$ .

The Uniqueness Theorem of the power series.

Thm. Suppose that the function of power series

$$f: (-R, R) \rightarrow \mathbb{R}: x \mapsto \sum_{n=0}^{\infty} a_n x^n$$

is well-defined.

If  $Z(f)$  has a limit point in  $(-R, R)$ , then  $a_n = 0$  for all  $n \geq 0$ .

Proof. Let  $A = \{x \in (-R, R) \mid x \text{ is the limit point of } Z(f)\}$ . ( $A$  is non-empty, by the hypo-)

Given  $x_0 \in A$ , the representation

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \text{ is valid in } |x - x_0| < R - |x_0|.$$

To show that  $a_n = 0$  for all  $n \geq 0$ , let  $k$  be the smallest non-negative integer s.t.  $a_k \neq 0$ .

Then, we can write  $f(x) = (x - x_0)^k \cdot g(x)$  where  $g(x) = \sum_{m=0}^{\infty} a_{k+m} (x - x_0)^m$ . ( $|x - x_0| < R - |x_0|$ ).

And, the  $g$  is continuous at  $x_0$ , because the each term is continuous and converges uniformly.

And,  $g(x_0) \neq 0$  by the construction. So, for some  $\delta > 0$ ,  $g(x) \neq 0$  in  $|x - x_0| < \delta$ .

So,  $f(x) \neq 0$  in  $0 < |x - x_0| < \delta$ , it contradicts to  $x_0$  is the limit point of  $Z(f)$ .

Therefore,  $f(x) = 0$  for all  $|x - x_0| < R - |x_0|$ , it means that

$$x_0 \in \overline{B_{R-|x_0|}(x_0)} \subseteq Z(f), \text{ and so } B_{R-|x_0|} \cap (-R, R) \subseteq A, \text{ so } A \text{ is open.}$$

And, the set of limit points is closed,  $A$  is the non-empty clopen set of  $(-R, R)$ .

But, since  $(-R, R)$  is connected, the only possibility is  $A = (-R, R)$ .

Moreover, since  $f$  is continuous,  $A \subseteq Z(f)$ , so  $Z(f) = (-R, R)$ .  $\square$